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Volatility Surface Modelling: Black-Scholes, Dupire, and Heston on FTSE 100 Options

March 2026

Introduction

Volatility plays a key role in financial markets, influencing asset valuation, hedging decisions, and risk management. While the classical Black–Scholes model [Black and Scholes 1973] assumes constant volatility, empirical evidence shows that market volatility varies across time and the given state of the market. Over time, this motivated the development of more complex volatility models, such as Dupire's local volatility model [Dupire 1994], which models volatility as a deterministic function of asset price and time, and Heston's stochastic volatility model [Heston 1993], which captures more realistic dynamics by modelling volatility as its own stochastic process. These models provide alternative ways to view volatility behaviour.

This paper compares these three approaches, constant, deterministic, and stochastic volatility, to understand how each represents uncertainty and how these differences affect options pricing and risk management. Key to interpreting the investment risks, implications for hedging strategies, and calibration stability, is understanding the nature and mathematical structures which govern each model. The nature of these partial differential equations, and stochastic calculus, enables us to gain insights into robust hedging and capital decisions.

To compare models, statistical analysis was conducted of one complete days (2nd January 2026) of actual historical data, accessed from the Bloomberg terminal, of the FTSE 100 index option chain. The FTSE 100 index was chosen as each model was developed as an analytical solution to European-style options. The FTSE 100 index is a highly liquid equity index with a deep options market which results in a rich dataset.

Using Python, the data was collated, from which the implied volatilities were extracted for the Black-Scholes model using the numerical method of Newton-Raphson, which iterated around the difference in the market price compared to the theoretical Black-Scholes price. Subsequently, the local volatilities were calibrated for the Dupire model and plotted. To compute the Heston prices, complex intergals were analysed and computed through rectangular integration, and to find the unknown parameters, the method of ordinary least squared errors was used. To find the Heston implied volatilities, the difference between the Heston price and the theoretical Black-Scholes price is minimised. Monte-Carlo simulations are utilised to analyse different asset price and variance paths. Distributional skews are plotted and Value at Risk (VaR) and Expected Shortfall (ES) estimates are calculated. The objective is to compare not only pricing accuracy but also how effectively each model captures the shape of the volatility surface, handles different regions of moneyness and maturity, and performs under realistic market conditions for European options.

General Concepts

A financial option is a financial derivative which represents a contract between the buyer (holder) and the seller (writer). The buyer has the right, but not the obligation, to buy or sell an underlying asset (*e.g.*, stocks, currencies, or commodities) at a predetermined price (strike price) within a specific time frame. The holder pays a fee, known as a premium, to the seller for this right.

2.1 Why Use an Option Pricing Model?

Each underlying asset has multiple variables, with many possible options, for example, differences in strike price, maturity, and premium. A model allows us to reduce multiple variables to one common variable, which is implied volatility. This way, we can express an underlying asset in a common language. It puts all options on the same scale, allowing us to compare contracts consistently and identify which options have the highest probability of finishing in the money.

2.2 Introduction to Volatility

Volatility reflects the uncertainty in the underlying asset's price, mathematically it is the standard deviation of the underlying asset's return, represented by σ. Thus, the higher it is, the greater the potential for options to fluctuate in price, this means that there is a higher chance of finishing in the money. Thus, higher volatility will increase options prices.

In Figure (1), we have a visual representation that the same option with a strike price above E_1 and a current asset price below E_1 will be priced higher when its volatility is higher since it has a greater chance of expiring above E_1, represented by a larger area under the curve. The Black-Scholes equation assumes volatility to be

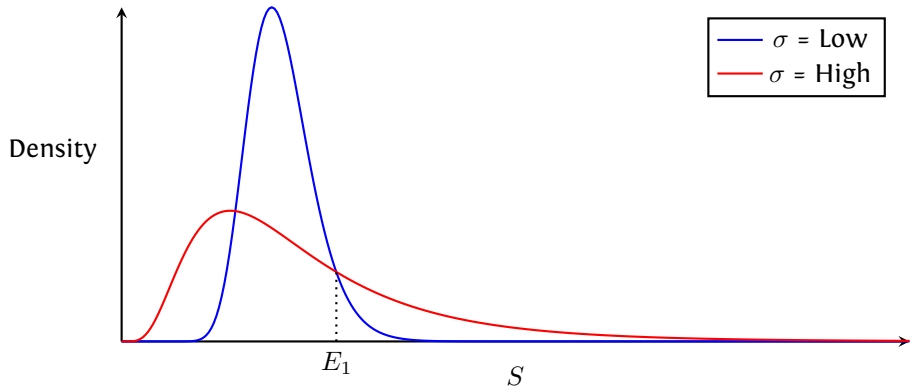

Figure 1: Lognormal distributions with low and high volatility

constant, which in practice, is not true. We can use volatility models, such as the Dupire and Heston models to improve on this limitation.